

All Dice, No Slice: Metric Artifacts, Data Leakage, and Task Interference in Multi-Task Computational Pathology

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Abstract

Simultaneous classification and lesion segmentation via hard parameter-sharing U-Nets is a compact way to couple both objectives in computational pathology. Recently, classification accuracies of 82% – 90% and Dice scores of 97% – 99% were reported across diverse medical imaging modalities using standard convolutional backbones and static loss weighting. We conduct a 26-experiment replication and methodological audit across four datasets: TCGA-LGG brain-tumor MRI, PANDA prostate biopsies, SIIM-ACR pneumothorax radiographs, and the 19-tissue PanNuke benchmark. Under strict patient-level data isolation (**GroupKFold**), the claimed accuracies are not reproducible, and the reported Dice levels are attainable only as empty-mask artifacts: all four of our SIIM configurations converge to all-empty predictions yet report 77.74% Dice — exactly the lesion-free fraction of the validation split under the convention that scores empty-over-empty slices as perfect agreement. On 6-class ordinal prostate cancer grading (PANDA), empirical top-1 accuracy spans 29.04% – 46.15% and Dice spans 28.94% – 44.34% across all 12 evaluated configurations, with static 5:1-weighted 10^{-3} baselines (Runs 03–04) reaching 45.15% – 46.15% / 43.30% – 44.34% (vs. 88.0% and 99.0% claimed). Three systemic causes explain the discrepancy: (1) empty-mask Dice crediting, (2) patient-level

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data leakage, and (3) task interference from the joint optimization recipe. Matched ablations further show that skip connections are a null result, whereas removing Macenko stain normalization *improves* subtype discrimination (+5.51% on PANDA, +2.68% on PanNuke). Our benchmark and audit protocol are released to prevent such metric artifacts in future multi-task pathology studies.

Keywords: Computational Pathology, Multi-Task Learning, U-Net, Reproducibility, Metric Artifacts, Data Leakage, GradNorm

1. Introduction

Deep learning systems in digital pathology and radiology are increasingly required to deliver both diagnostic categorization and lesion delineation [1, 2]. Large visual foundation models (e.g., UNI [3], CONCH [4], and Virchow [5]) show that clinically actionable representations require massive pretraining and self-supervised objectives across gigapixel whole-slide images (WSIs). In resource-constrained deployments, hard parameter sharing via an encoder-decoder backbone (a U-Net [6] with dual prediction heads) remains a lightweight alternative that couples both objectives in a single forward pass.

Recently, Rhanoui et al. [7] claimed that a standard U-Net with lightweight encoders (VGG16 [8] and MobileNetV2 [9]) trained with static loss weights ($\lambda_{seg} = 5.0$, $\lambda_{cls} = 1.0$) and a high learning rate ($\eta = 10^{-3}$) simultaneously solves both classification and segmentation across four distinct medical modalities (Brain MRI, Skin Dermoscopy, Prostate Histopathology, and Chest X-Ray), reporting 82% – 90% classification accuracy and 97% – 99% Dice similarity coefficients.

In this paper, we report a 26-run replication study and systematic methodological audit. Our findings demonstrate that:

1. **Empirical Irreproducibility:** When evaluated with strict patient/biopsy-level isolation (**GroupKFold**), multi-class grading on PANDA prostate biopsies degrades from the claimed 88.0% accuracy / 99.0% Dice to an empirical range of 29.04% – 46.15% Acc / 28.94% – 44.34% Dice across all 12 configurations, with the static 5:1-weighted 10^{-3} baselines (Runs 03–04) achieving 45.15% – 46.15% / 43.30% – 44.34%.
2. **The Empty-Mask Dice Metric Artifact:** Near-perfect (99%) Dice claims on sparse-lesion datasets such as SIIM-ACR pneumothorax can

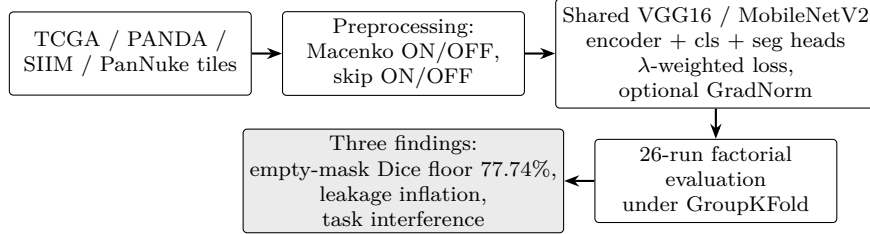


Figure 1: Overview of the 26-run replication and audit pipeline. Four source datasets (TCGA-LGG, PANDA, SIIM-ACR, PanNuke) are tiled and passed through a preprocessing stage that toggles Macenko stain normalization and skip connections. A shared VGG16/MobileNetV2 encoder feeds a classification head and a segmentation head trained with a λ -weighted loss (static ratios or optional GRADNORM). All 26 configurations are evaluated under patient-level GroupKFold isolation, yielding three findings: the empty-mask Dice floor of 77.74% on SIIM, patient-level leakage inflation of classification accuracy, and task interference from the joint optimization recipe.

arise from crediting lesion-free slices ($|Y| = |\hat{Y}| = 0$) as Dice = 1.0; our four SIIM runs measure exactly this floor (77.74%) under an all-empty predictor.

3. **Controlled Multi-Task & Ablation Analysis:** Through matched factorial controls (learning-rate pairs, static loss-weight ratios $\lambda_{seg} : \lambda_{cls} \in \{1:10, 1:1, 5:1, 10:1\}$, GRADNORM [10] balancing, Macenko normalization, and skip-connection ablations), we isolate how individual components govern multi-task optimization: GradNorm-style balancing actively impairs 6-class ordinal grading (45.39% $\rightarrow$ 29.04% Acc), while skip connections show no resolvable difference.

To facilitate complete reproducibility, all training scripts, raw JSON summaries for all 26 runs, evaluation checkpoints, and generated figures are publicly available at <https://github.com/ApatheticMioz/cancer-pathology-dl> [11]. Figure 1 summarizes the end-to-end pipeline and the three findings it yields.

2. Multi-Task Formulation & Evaluation Metrics

2.1. Architectural Formulation

Let $\mathbf{x} \in \mathbb{R}^{H \times W \times C}$ denote an input image tile. A shared convolutional encoder $f_\theta(\mathbf{x})$ extracts a bottleneck latent representation $\mathbf{z} \in \mathbb{R}^{h \times w \times d}$. The

classification branch applies global average pooling (GAP) followed by a two-layer multilayer perceptron (hidden width 256, dropout 0.5):

$$\hat{\mathbf{y}}_{cls} = \text{Softmax}(\mathbf{W}_2 \phi(\mathbf{W}_1 \text{GAP}(\mathbf{z}) + \mathbf{b}_1) + \mathbf{b}_2) \in \Delta^{K-1}, \quad (1)$$

where K is the number of diagnostic classes, Δ^{K-1} denotes the probability simplex, and $\phi(\cdot)$ is the ReLU nonlinearity. The segmentation branch passes $\mathbf{z}$ through a decoder Dec_ϕ with lateral skip connections $\{\mathbf{s}_l\}_{l=1}^L$ from encoder stages, yielding unnormalized pixel logits $\mathbf{l}_{seg} \in \mathbb{R}^{H \times W \times M}$. For binary segmentation ($M = 1$), $\hat{\mathbf{y}}_{seg} = \sigma(\mathbf{l}_{seg})$ with $\sigma(\cdot)$ the sigmoid; for $M > 1$, class probabilities are obtained via channel-wise Softmax.

The joint multi-task training objective is defined as:

$$\mathcal{L}_{total}(\theta, \phi, \mathbf{W}_{cls}) = \lambda_{cls} \mathcal{L}_{cls}(\mathbf{y}_{cls}, \hat{\mathbf{y}}_{cls}) + \lambda_{seg} \mathcal{L}_{seg}(\mathbf{y}_{seg}, \mathbf{l}_{seg}), \quad (2)$$

where $\mathcal{L}_{cls}$ is categorical cross-entropy $\mathcal{L}_{CE}$. In the published study of Rhanoui et al. [7], the segmentation loss was stated as $\mathcal{L}_{seg} = \mathcal{L}_{BCE} + \mathcal{L}_{Dice}$; the implementation audited for this study, however, optimizes purely cross-entropy objectives, without any Dice-based penalty term (a documented deviation):

$$\mathcal{L}_{seg}(\mathbf{y}_{seg}, \mathbf{l}_{seg}) = \begin{cases} \mathcal{L}_{BCE}(\mathbf{y}_{seg}, \mathbf{l}_{seg}), & M = 1 \text{ (binary tasks)}, \\ \mathcal{L}_{CE}(\mathbf{y}_{seg}, \mathbf{l}_{seg}), & M > 1 \text{ (multi-class tasks)}. \end{cases} \quad (3)$$

The Dice similarity coefficient (DSC) is evaluated post-hoc as a validation metric, not as an active loss term during gradient descent.

2.2. The Empty-Mask Dice Metric Artifact

The Sørensen–Dice coefficient between ground-truth mask Y and binary prediction $\hat{Y}$ ($Y, \hat{Y} \in \{0, 1\}^{H \times W}$) is:

$$\text{Dice}(Y, \hat{Y}) = \frac{2|Y \cap \hat{Y}|}{|Y| + |\hat{Y}| + \epsilon}, \quad \epsilon = 10^{-8}; \quad \text{Dice} := 1 \text{ if } |Y| + |\hat{Y}| = 0. \quad (4)$$

Empty references and predictions are a well-recognized validation pitfall in biomedical image analysis [12]. Two conventions govern empty slices:

- **Foreground-Only Evaluation:** Compute Dice strictly over non-empty target slices ($\{i \mid |Y_i| > 0\}$).

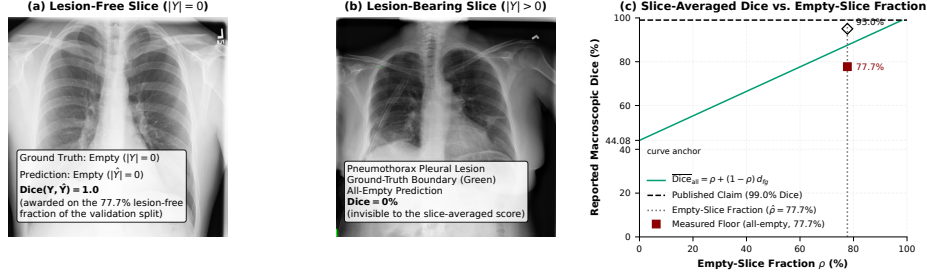


Figure 2: The empty-mask Dice artifact on SIIM-ACR pneumothorax. (a) Lesion-free slice ($|Y| = 0$): under empty-credit assignment an all-empty prediction is awarded Dice = 1.0; lesion-free slices constitute $\hat{\rho} = 77.7\%$ of our validation split. (b) Lesion-bearing slice ($|Y| > 0$, ground-truth boundary in green): our all-empty models predict empty masks here and score Dice = 0, invisible to the slice-averaged mean. (c) Convex-combination curve (Eq. 6): a foreground Dice of 77.74% would report 95.04% at $\hat{\rho} = 0.777$ (analytic illustration), whereas all four of our SIIM runs measure the all-empty floor of 77.74% itself.

- **Empty-Credit Assignment:** Assign Dice = 1.0 when $|Y| = 0 \wedge |\hat{Y}| = 0$, and Dice = 0.0 when $|Y| = 0 \wedge |\hat{Y}| > 0$.

Averaged over all slices under empty-credit assignment, the macroscopic mean is

$$\overline{\text{Dice}}_{\text{all}} = \frac{1}{N} \sum_{i=1}^N d_i, \quad d_i = \text{Dice}(Y_i, \hat{Y}_i) \text{ if } |Y_i| > 0, \text{ else } \mathbf{1}[\hat{Y}_i = 0], \quad (5)$$

which for mean foreground Dice $\overline{\text{Dice}}_{\text{fg}}$ on lesion-bearing slices is the convex combination

$$\overline{\text{Dice}}_{\text{all}} = \rho + (1 - \rho) \cdot \overline{\text{Dice}}_{\text{fg}}, \quad (6)$$

where ρ is the proportion of lesion-free slices. In sparse pathologies such as SIIM-ACR pneumothorax, a model with a genuine foreground Dice of 77.74% would report $\overline{\text{Dice}}_{\text{all}} = 0.777 \times 1.0 + 0.223 \times 0.7774 = 95.04\%$ at the observed lesion-free fraction $\hat{\rho} = 1,659/2,135 = 0.777$ (an analytic illustration). Our four SIIM runs realize a more extreme failure mode. They emit empty masks on every validation slice throughout training, and each reported 77.74% (Runs 05, 06, 11, 12) *is the all-empty floor itself* — Eq. 5 with zero foreground Dice. Fig. 2 shows why this matters: on sparse data, slice-averaged Dice reports the lesion-free fraction rather than lesion overlap. Published 99.0% Dice claims [7] are therefore uninterpretable as evidence of lesion localization.

2.3. Adaptive Gradient Balancing via GradNorm

To test whether dynamic loss balancing mitigates cross-task gradient competition, we introduced a parameterized variant of GRADNORM [10] in Phase v2.

Let W denote the shared convolutional filters and normalization layers of the U-Net encoder f_θ . The L_2 gradient norm of task $i \in \{seg, cls\}$ with respect to W at minibatch t is:

$$G_W^{(i)}(t) = \|\nabla_W w_i(t) \mathcal{L}_i(t)\|_2. \quad (7)$$

For stability in the primary Adam optimizer, balancing weights are learnable log-weights $\mu_i(t) = \ln w_i(t)$ (initialized to $w_{seg}(0) = 5.0$, $w_{cls}(0) = 1.0$), updated jointly with the network parameters. Minibatch optimization minimizes the L_1 gradient disparity objective:

$$\mathcal{L}_{grad}(t) = \sum_{i \in \{seg, cls\}} \left| G_W^{(i)}(t) - \bar{G}_W(t) \cdot [r_i(t)]^\alpha \right|, \quad (8)$$

where $\bar{G}_W(t) = \frac{1}{2} \sum_j G_W^{(j)}(t)$ is the average gradient norm across tasks, $\alpha = 1.5$ is the asymmetry parameter, and $r_i(t)$ represents the relative inverse training rate:

$$r_i(t) = \frac{\tilde{\mathcal{L}}_i(t)}{\frac{1}{2} \sum_{j \in \{seg, cls\}} \tilde{\mathcal{L}}_j(t)}, \quad \text{with} \quad \tilde{\mathcal{L}}_i(t) = \frac{\mathcal{L}_i(t)}{\mathcal{L}_i(0)}, \quad (9)$$

where $\mathcal{L}_i(0)$ is the initial step-0 loss. Following Chen et al. [10], slower-training tasks exhibit higher loss ratios $r_i(t) > 1$, which raises their target gradient norm $\bar{G}_W(t)[r_i(t)]^\alpha$. After each optimizer update, we renormalize the weights so that $\sum_i w_i(t) = 2$. In the canonical formulation [10], the module differentiates $\mathcal{L}_{grad}$ against detached targets and applies it to the balancing weights alone through a dedicated optimizer (learning rate 0.025). §4.3 probes this distinction empirically. We evaluate the module across Groups 2 and 4 to isolate how dynamic gradient equalization behaves in hard-parameter-sharing architectures.

2.4. Color Deconvolution and Optical Density Space

To mitigate staining variability across clinical laboratories, the v2 pipeline incorporates Macenko stain normalization [13]. We convert RGB pixel intensities $\mathbf{I} \in [0, 255]^3$ to optical density (OD) space:

$$\mathbf{OD} = -\ln \left(\frac{\mathbf{I} + \epsilon}{I_0} \right), \quad (10)$$

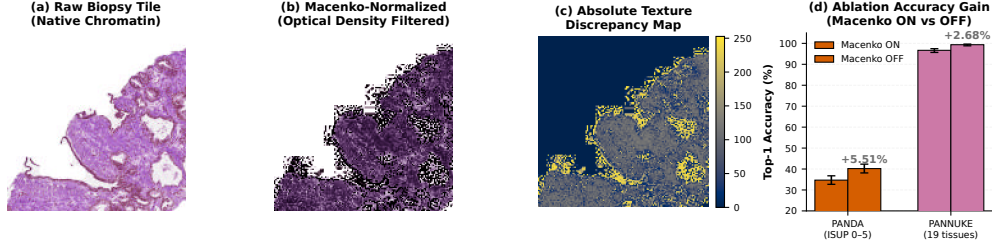


Figure 3: Visual and quantitative impact of Macenko stain normalization on histopathology classification. (a) Raw prostate biopsy tile. (b) Optical-density-projected tile: color deconvolution acts as a chrominance low-pass filter, flattening sub-cellular texture. (c) Absolute texture discrepancy map. (d) Empirical top-1 accuracy gains across matched models when Macenko normalization is bypassed (+5.51% on PANDA, +2.68% on PanNuke).

where $I_0 = 255$ represents transmitted light intensity. The implementation computes optical density via natural logarithms $-\ln(\cdot)$ rather than the canonical base-10 formulation $-\log_{10}(\cdot)$ [13]. We estimate the hematoxylin and eosin stain vectors by eigendecomposition of the optical-density covariance matrix (stain-vector angles restricted to the 1st–99th percentile range) over OD values above a threshold of 0.15, and project them onto a target reference vector. This step effectively eliminates macroscopic batch variations. It also acts as a chrominance low-pass filter, however: projecting raw RGB channels onto static stain vectors smooths the sub-cellular chromatin textures that are clinically essential for grading fine glandular architectures (Fig. 3).

3. The 26-Run Empirical Reproduction Campaign

3.1. Datasets and Experimental Protocol

We evaluated four benchmark datasets:

1. **TCGA-LGG Brain Tumor:** 2-class lesion classification and FLAIR abnormality segmentation on 2D 256×256 slices ($N = 3,929$; 110 patients [14]).
2. **PANDA Prostate Biopsies:** 6-class ISUP Gleason grading and gland segmentation ($N = 10,616$ tiles [15]; patient-level GroupKFold isolation).
3. **SIIM-ACR Pneumothorax:** Binary chest-radiography detection and pleural boundary segmentation ($N = 12,047$ images [16]).

Table 1: 26-run experimental results matrix across all five evaluation groups. V1 = original static baseline ($\eta = 10^{-3}$, $\lambda_{seg} : \lambda_{cls} = 5 : 1$, raw input); V2 = GradNorm variant ($\alpha = 1.5$, $\eta = 10^{-4}$, Macenko normalization). $\checkmark$ = GradNorm active; — = not used. Markers: $\times$ = skip connections zeroed; $\diamond$ = raw input without Macenko normalization (v1 runs are raw by construction). Rows 17–22 (Group 4) systematically isolate learning rate ($\eta = 10^{-4}$ vs 10^{-3}), GradNorm ($\alpha = 1.5$, Run 18), and loss weighting ratios on PANDA $\times$ VGG16 under raw input. Pub. Acc/Dice: published reference targets of Rhanoui et al. [7] (— = no published target). Δ_{Acc} / Δ_{Dice} : difference relative to published targets. All metrics in %. Complete 95% Wilson score intervals for all 26 configurations are tabulated in the open-source repository matrix (`paper_results_matrix_with_ci.csv`).

#	Configuration	Dataset	Encoder	LR	η	GN	Val	Acc	Val	Dice	Pub.	Acc	Pub.	Dice	Δ_{Acc}	Δ_{Dice}
Group 1: Baseline Reproductions (static 5:1, $\eta = 10^{-3}$, raw input)																
01	Naked Baseline	TCGA	VGG16	10^{-3}	—	—	88.82	76.02	89.0	97.0	—	97.0	—	—	−0.18	−20.98
02	Naked Baseline	TCGA	MobileNetV2	10^{-3}	—	—	94.60	87.56	90.0	98.0	—	98.0	—	—	+4.60	−10.44
03	Naked Baseline	PANDA	VGG16	10^{-3}	—	—	45.15	43.30	87.0	98.0	—	98.0	—	—	−41.85	−54.70
04	Naked Baseline	PANDA	MobileNetV2	10^{-3}	—	—	46.15	44.34	88.0	99.0	—	99.0	—	—	−41.85	−54.66
05	Naked Baseline	SIIM	VGG16	10^{-3}	—	—	83.19	77.74	82.0	99.0	—	99.0	—	—	+1.19	−21.26
06	Naked Baseline	SIIM	MobileNetV2	10^{-3}	—	—	83.47	77.74	87.0	99.0	—	99.0	—	—	−3.53	−21.26
Group 2: V2 Package (GradNorm $\alpha = 1.5$, $\eta = 10^{-4}$, Macenko normalization)																
07	GradNorm v2	TCGA	VGG16	10^{-4}	$\checkmark$	$\checkmark$	92.80	78.30	89.0	97.0	—	97.0	—	—	+3.80	−18.70
08	GradNorm v2	TCGA	MobileNetV2	10^{-4}	$\checkmark$	$\checkmark$	93.32	84.20	90.0	98.0	—	98.0	—	—	+3.32	−13.80
09	GradNorm v2	PANDA	VGG16	10^{-4}	$\checkmark$	$\checkmark$	30.89	31.59	87.0	98.0	—	98.0	—	—	−56.11	−66.41
10	GradNorm v2	PANDA	MobileNetV2	10^{-4}	$\checkmark$	$\checkmark$	34.70	31.34	88.0	99.0	—	99.0	—	—	−53.30	−67.66
11	GradNorm v2	SIIM	VGG16	10^{-4}	$\checkmark$	$\checkmark$	81.08	77.74	82.0	99.0	—	99.0	—	—	−0.92	−21.26
12	GradNorm v2	SIIM	MobileNetV2	10^{-4}	$\checkmark$	$\checkmark$	83.00	77.74	87.0	99.0	—	99.0	—	—	−4.00	−21.26
Group 3: PanNuke 19-Tissue Multi-Organ Benchmark (external control; no published targets)																
13	Naked Baseline	PanNuke	VGG16	10^{-3}	—	—	98.85	72.48	—	—	—	—	—	—	—	—
14	Naked Baseline	PanNuke	MobileNetV2	10^{-3}	—	—	99.43	74.21	—	—	—	—	—	—	—	—
15	GradNorm v2	PanNuke	VGG16	10^{-4}	$\checkmark$	$\checkmark$	91.90	39.59	—	—	—	—	—	—	—	—
16	GradNorm v2	PanNuke	MobileNetV2	10^{-4}	$\checkmark$	$\checkmark$	96.68	60.54	—	—	—	—	—	—	—	—
Group 4: Optimization Isolation (PANDA $\times$ VGG16: LR, GradNorm, loss weighting; raw input)																
17	Isolate LR: static $\eta = 10^{-4}$ $\diamond$	PANDA	VGG16	10^{-4}	—	—	43.35	37.99	87.0	98.0	—	98.0	—	—	−43.65	−60.01
18	Isolate GradNorm: $\alpha = 1.5$	PANDA	VGG16	10^{-3}	$\checkmark$	$\checkmark$	29.04	31.43	87.0	98.0	—	98.0	—	—	−57.96	−66.57
19	Loss weight $\lambda = 1:1$	PANDA	VGG16	10^{-3}	—	—	42.73	41.72	87.0	98.0	—	98.0	—	—	−44.27	−56.28
20	Loss weight $\lambda = 5:1$	PANDA	VGG16	10^{-3}	—	—	45.39	44.08	87.0	98.0	—	98.0	—	—	−41.61	−53.92
21	Loss weight $\lambda = 1:10$	PANDA	VGG16	10^{-3}	—	—	41.83	38.06	87.0	98.0	—	98.0	—	—	−45.17	−59.94
22	Loss weight $\lambda = 10:1$	PANDA	VGG16	10^{-3}	—	—	44.58	41.52	87.0	98.0	—	98.0	—	—	−42.42	−56.48
Group 5: Ablation Matrix (stain normalization $\diamond$ and skip connections $\times$)																
23	Ablation: No-Macenko $\diamond$	PANDA	MobileNetV2	10^{-4}	$\checkmark$	$\checkmark$	40.21	31.57	88.0	99.0	—	99.0	—	—	−47.79	−67.43
24	Ablation: No-Macenko $\diamond$	PanNuke	MobileNetV2	10^{-4}	$\checkmark$	$\checkmark$	99.36	73.33	—	—	—	—	—	—	—	—
25	Ablation: No-Skips $\times$	TCGA	MobileNetV2	10^{-4}	$\checkmark$	$\checkmark$	94.34	84.12	90.0	98.0	—	98.0	—	—	+4.34	−13.88
26	Ablation: No-Skips $\times$	PANDA	MobileNetV2	10^{-4}	$\checkmark$	$\checkmark$	35.31	28.94	88.0	99.0	—	99.0	—	—	−52.69	−70.06

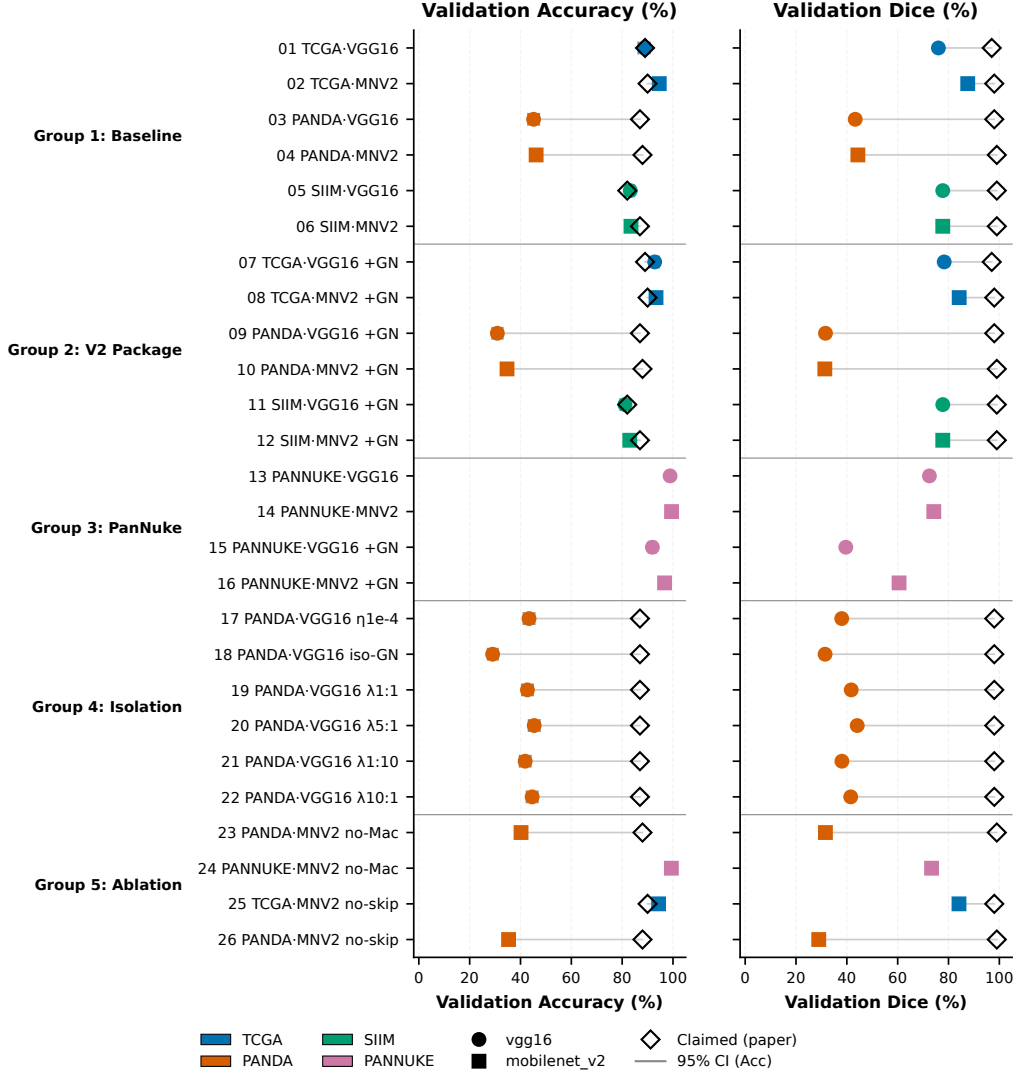


Figure 4: 26-run claimed-vs-measured dot plot. Left panel: validation accuracy (%) with 95% Wilson-CI whiskers (measured, filled markers) and the published claimed accuracy (black open diamond); the tie line between them is the gap that constitutes the paper’s central argument. Right panel: validation Dice (%) with the same layout but no whiskers, as no published confidence intervals exist for the macro-Dice claims. The PANDA rows (Runs 03–04, 09–10, 17–22, 23, 26) show the largest gaps: measured accuracy spans 29.04%–46.15% vs. claimed 87–88%, and Dice spans 28.94%–44.34% vs. claimed 98–99%. All four SIIM rows (Runs 05, 06, 11, 12) measure exactly 77.74% Dice — the all-empty floor. PanNuke rows (Runs 13–16, 24) have no published targets and are shown without claimed diamonds.

4. **PanNuke:** 19-tissue multi-organ benchmark ($N = 7,901$ patches [17]); having no published multi-task targets, it serves as an unclaimed external control.

We trained all models with PyTorch (up to 50 epochs with early stopping, patience 10; batch size 32; fixed seed 42) using ImageNet-pretrained encoders and the Adam optimizer with a constant per-phase learning rate and zero weight decay, and we applied no learning-rate schedule or additional regularization. In Phase v1 (Runs 01–06), we replicated the original static setup ($\eta = 10^{-3}$, $\lambda_{seg} : \lambda_{cls} = 5 : 1$, raw inputs). In Phase v2 (Runs 07–12), we incorporated dynamic GRADNORM ($\alpha = 1.5$, $\eta = 10^{-4}$) and Macenko stain normalization [13]. Groups 3–5 (Runs 13–26) systematically evaluate multi-organ generalizability, optimization hyperparameter isolation, and architectural skip ablations. Table 1 summarizes all 26 runs and Figure 4 visualizes them.

4. Discussion & Methodological Anatomy

4.1. The Anatomy of the PANDA Collapse

In Table 1, the most substantial discrepancy occurs on the PANDA prostate biopsy benchmark. The original authors reported 88.0% accuracy on 6-class ISUP grading alongside a 99.0% Dice score. In contrast, across all 12 PANDA configurations tested, empirical top-1 accuracy spans 29.04% – 46.15% and Dice spans 28.94% – 44.34%. The static 5:1-weighted 10^{-3} baselines (Runs 03–04) achieve 45.15% / 43.30% (VGG16) and 46.15% / 44.34% (MobileNetV2). Run 04 is the strongest single configuration in our campaign.

This outcome is consistent with established clinical and machine learning benchmarks. In the international PANDA challenge (1,010 participating teams) [15], the top-performing whole-slide multi-instance learning algorithms reached external validation quadratic weighted κ of 0.862 (95% CI 0.840–0.884) and 0.868 (0.835–0.900) on U.S. and European cohorts, and inter-observer Gleason grading of biopsy cores is itself imperfect ($\kappa = 0.68$ among urologic pathologists and 0.435 among general pathologists [18, 19]). Claiming 88.0% accuracy on 128×128 isolated patches with a standard 2D CNN is not attainable without patient-level data leakage: tiles from the same biopsy core contaminate both train and test splits. Figure 5 traces the PANDA accuracy drop across training epochs and isolates the responsible mechanism.

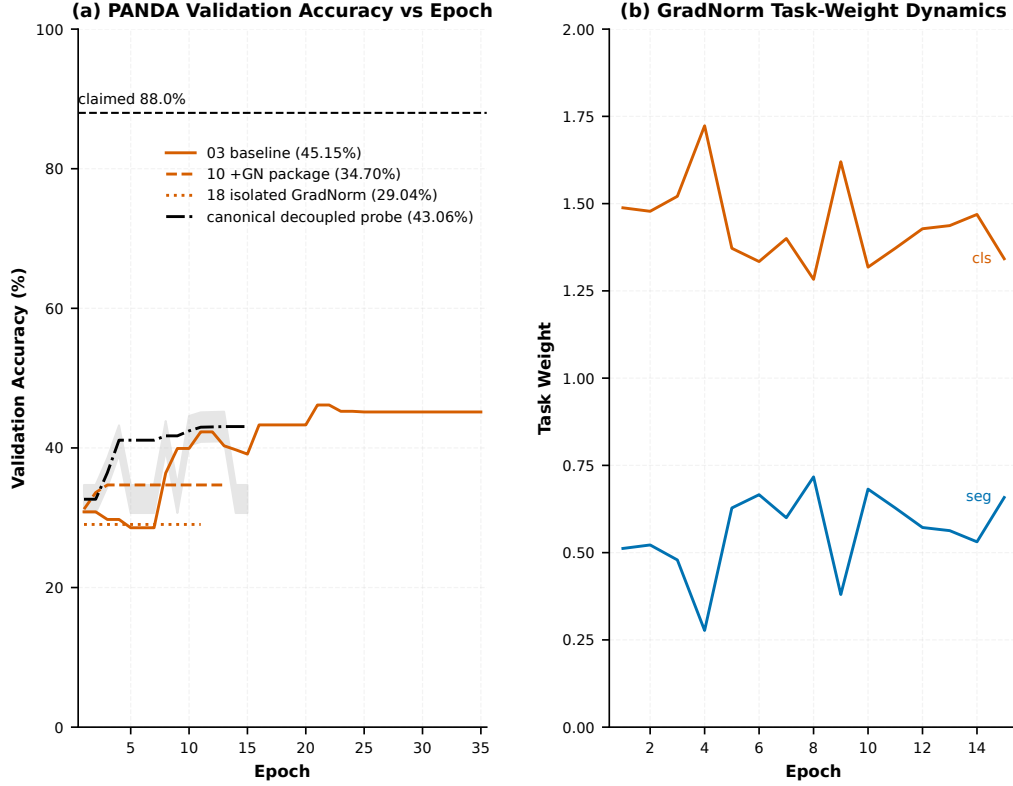


Figure 5: Anatomy of the PANDA accuracy drop. (a) Validation accuracy vs. epoch for three PANDA runs — Run 03 (static 5:1 baseline, 45.15%), Run 10 (v2 GradNorm package, 34.70%), Run 18 (isolated GradNorm at $\eta = 10^{-3}$, 29.04%) — plus the canonical decoupled GRADNORM probe (dash-dot, shaded 95% CI band), which remains stable but plateaus at 43.06% over 15 epochs, 44.94 percentage points below the claimed 88.0% (black dashed reference). (b) GRADNORM task-weight dynamics from the canonical log: the segmentation and classification weights over 15 epochs, showing the dynamic rebalancing that the joint variant turns into destructive updates on the shared encoder.

4.2. Controlled Skip Connection Ablation: A Verified Null Result

Skip connections are evaluated in matched pairs (learning rate, GradNorm, Macenko normalization held identical):

- **Coarse macro-tumor structures (TCGA):** Run 25 (skip-ablated) vs. matched baseline Run 08 (both $\eta = 10^{-4}$, GradNorm, Macenko): accuracy 93.32% $\rightarrow$ 94.34% ($\Delta_{\text{Acc}} = +1.02$), Dice 84.20% vs 84.12% ($\Delta_{\text{Dice}} = -0.08$).
- **Fine micro-glandular structures (PANDA):** Run 10 (34.70% Acc / 31.34% Dice) vs. matched skip-ablated Run 26 (35.31% / 28.94%): $\Delta_{\text{Acc}} = +0.61$, $\Delta_{\text{Dice}} = -2.40$.

In both cases the differences are small, and the 95% Wilson score intervals on top-1 accuracy overlap substantially (TCGA: [91.34, 94.87] vs [92.49, 95.76]; PANDA: [32.69, 36.76] vs [33.30, 37.38]). We therefore report these as *verified null results*: the bottleneck-only representation suffices for both coarse macro-lesion and fine micro-glandular segmentation; zeroing the skips produces no detectable penalty in a single-run comparison. A definitive directional claim would require multi-seed replication of the matched pairs.

Macenko stain-normalization ablation.. The matched Macenko pair (Group 5, Runs 23–24) is the only single-run ablation with a statistically resolvable classification shift: omitting normalization raises PANDA top-1 accuracy by +5.51 and PanNuke accuracy by +2.68 (disjoint 95% CIs), while Dice stays flat on PANDA and rises by +12.79 on PanNuke. Color deconvolution thus harms fine-grained classification, consistent with over-smoothing of the chromatin/texture cues on which grading relies, and partially decomposes the v1-vs-v2 PANDA gap of §4.1. Figure 6 consolidates all six matched pairs; Figure 3 shows the texture flattening.

4.3. Isolating Learning Rate from Dynamic Gradient Balancing

Group 4 (PANDA $\times$ VGG16) factorizes the key optimization hyperparameters: learning rate ($\eta \in \{10^{-3}, 10^{-4}\}$), GradNorm-style balancing (on/off), and static loss-weighting ratios $\lambda_{\text{seg}} : \lambda_{\text{cls}} \in \{1:10, 1:1, 5:1, 10:1\}$, all under raw inputs.

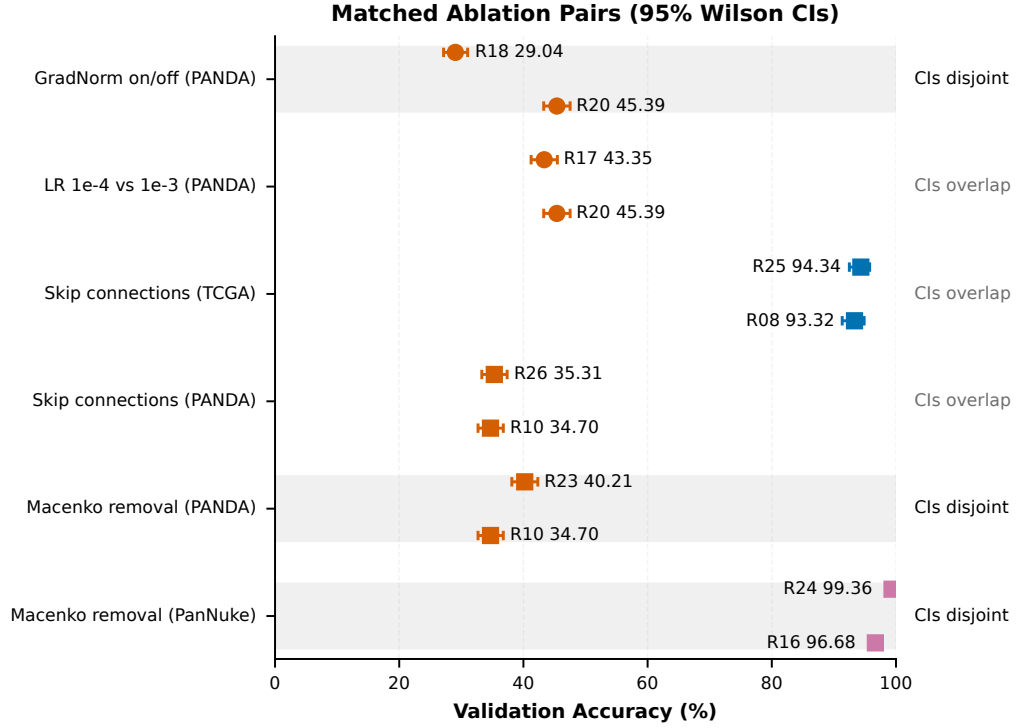


Figure 6: Matched-ablation forest plot of validation accuracy with 95% Wilson CIs for the six strictly matched pairs. Disjoint-CI pairs (shaded): isolated GradNorm (Run 18, 29.04%) vs. static 5:1 (Run 20, 45.39%); no-Macenko PANDA (Run 23, 40.21%) vs. matched (Run 10, 34.70%); no-Macenko PanNuke (Run 24, 99.36%) vs. matched (Run 16, 96.68%). Overlapping-CI pairs: LR 10^{-4} vs. 10^{-3} (Runs 17 vs. 20: 43.35% vs. 45.39%); skip ablation TCGA (Runs 25 vs. 08: 94.34% vs. 93.32%); skip ablation PANDA (Runs 26 vs. 10: 35.31% vs. 34.70%). The right-margin verdict column encodes the CI-overlap test, so the null-result status of the skip ablations and the resolvable Macenko/GradNorm effects are read directly from the plot.

Direct Isolation of Dynamic Gradient Balancing (Run 18)... The isolated GradNorm arm at $\eta = 10^{-3}$ under matched raw-input conditions (Run 18: PANDA $\times$ VGG16, GradNorm $\alpha = 1.5$, raw input) compares directly against the static 5:1 baseline (Run 20). Activating GradNorm ($\alpha = 1.5$) causes validation accuracy to collapse from 45.39% down to 29.04% (95% Wilson CI: [27.14, 31.02]), alongside a corresponding drop in segmentation Dice from 44.08% down to 31.43% ($\Delta_{\text{Acc}} = -16.35\%$, $\Delta_{\text{Dice}} = -12.65\%$, with non-overlapping confidence intervals: [43.27, 47.52] vs [27.14, 31.02]).

This 29.04% accuracy and 31.43% Dice matches the performance observed in Phase v2 at $\eta = 10^{-4}$ (Run 09: 30.89% Acc / 31.59% Dice). The $\approx 30\%$ floor on 6-class PANDA grading is thus driven primarily by the $\alpha = 1.5$ balancing formulation, whose gradient-norm equalization forces destructive updates onto the shared encoder, rather than by the lower learning rate.

Learning Rate and Loss Weighting Dynamics... Holding GradNorm off, reducing the learning rate from $\eta = 10^{-3}$ (Run 20: 45.39% Acc) to $\eta = 10^{-4}$ (Run 17: 43.35%) yields a minor change within overlapping 95% CIs. The static loss-ratio sweep (Runs 19–22) confirms $\lambda = 5:1$ as the best static balance, yet no ratio approaches the claimed 88% (Figure 7). The isolated GradNorm arm (Run 18) drops to 29.04% against the 45.39% static baseline with non-overlapping CIs, and the canonical decoupled scheme plateaus at 43.06% — dynamic balancing in either formulation cannot bridge the gap (Figures 5, 6).

5. Conclusion & Recommendations

Across 26 experiments, the claimed 88% – 99% metrics did not reproduce under patient isolation. The residual gap stems from dataset difficulty and the joint optimization recipe: skip connections are a null result, while joint balancing ($\alpha = 1.5$) destabilizes classification (45.39% $\rightarrow$ 29.04% Acc). We recommend: (1) mandatory patient/biopsy-level `GroupKFold` splitting; (2) explicit reporting of the Dice evaluation convention, with foreground-only or zero-penalty scoring on sparse-lesion datasets; and (3) 95% Wilson score intervals on matched single-run ablation comparisons to distinguish genuine architectural effects from stochastic variance. We further recommend (4) validation on diverse multi-organ benchmarks such as PanNuke, and (5) systematic evaluation of multi-task balancing hyperparameters before clinical translation.

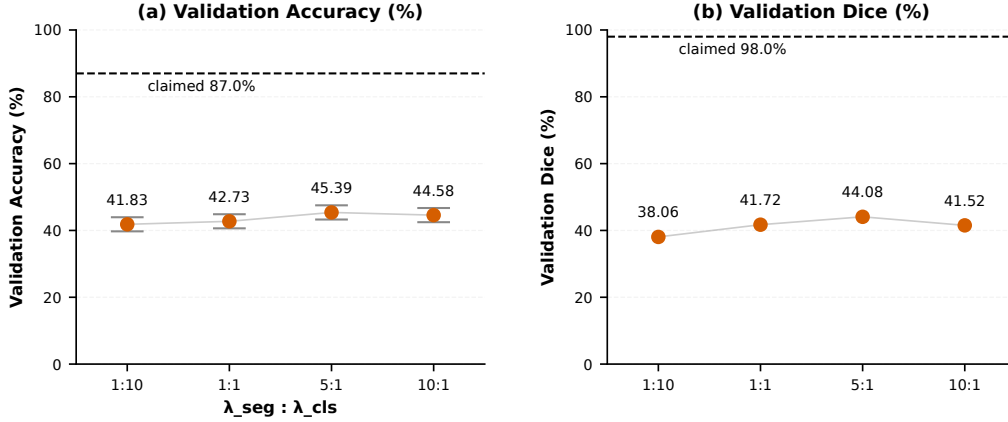


Figure 7: Static loss-weight ratio sweep on PANDA $\times$ VGG16 (raw input, $\eta = 10^{-3}$, no GradNorm). (a) Validation accuracy with 95% Wilson-CI whiskers: $\lambda_{seg} : \lambda_{cls} = 1:10$ (41.83%), 1:1 (42.73%), 5:1 (45.39%), 10:1 (44.58%); the black dashed line marks the claimed 87.0% accuracy. (b) Validation Dice for the same four arms (38.06%, 41.72%, 44.08%, 41.52%) against the claimed 98.0%. The 5:1 ratio is the best static balance on this architecture, yet no ratio approaches the claimed target — the gap is not a loss-weighting artifact.

Declaration of generative AI and AI-assisted technologies in the writing process

AI-assisted tools were used for language refinement and figure preparation. All scientific content, experiments, data analysis, and conclusions are the sole work of the authors, who reviewed and approved the final manuscript.

CRedit author statement

Muhammad Abdullah Ali: Conceptualization, Methodology, Software, Investigation, Writing – original draft. Muhammad Ibrahim Kiani: Supervision, Validation, Writing – review & editing. Muhammad Abdullah Aamir: Data curation, Visualization, Writing – review & editing.

Data availability

All training scripts, per-run JSON summaries, evaluation checkpoints, generated figures, and the complete results matrix with 95% Wilson score intervals (`paper_results_matrix_with_ci.csv`) are publicly available at <https://github.com/ApatheticMioz/cancer-pathology-dl>.

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